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ORIGINAL RESEARCH

Differences in Children's Physical Activity During Outdoor and Indoor School Recess in Hot Weather

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Background: Outdoor recess during school is a prominent source of children's physical activity; however, children's movement is likely affected when recess is moved indoors due to inclement weather. This study assessed the amount and intensity of children's physical activity during school recess when it took place inside gyms or classrooms compared with traditional outdoor recess during periods of high heat. ***Methods:*** This cross-sectional observational study included assessments of movement intensity levels during recess (sedentary, light, and moderate to vigorous physical activity [MVPA]) among fifth-grade children (N = 251; ages = 10–12 y) at 5 schools in the Phoenix Metropolitan Area (Arizona, United States) between August and September 2024. School personnel determined the locations for recess (in classrooms, gyms, or outdoors). Physical activity levels were assessed using the System for Observing Play and Leisure Activity in Youth during 91 observations across 15 recess periods. Differences in mean physical activity intensities were compared between recess locations for the entire sample and by child sex using 1-way analyses of variance. ***Results:*** Outdoor temperatures ranged from 91 to 109 °F (33–43 °C). Most recess was held outside (60%), with 20% in classrooms and 20% in gyms. All physical activity intensities differed between locations, with the greatest difference in time occurring in MVPA in gyms (51%) and outdoors (28%) compared with classrooms (11%), where children were mainly sedentary (71%). Boys spent more time in MVPA and less time sedentary than girls across all settings. ***Conclusions:*** Having indoor facilities to support movement such as gyms provides more MVPA and less time sitting compared with classrooms or outdoors on high-heat days.

Keywords: extreme heat, movement, climate, sedentary

Key Points

- This study examined how physical activity differed by recess location during extreme heat among fifth-grade students in Arizona elementary schools.
- Children accumulated substantially more time spent in moderate-to-vigorous physical activity during gym-based (51%) and outdoor (28%) recess compared to classroom recess (11%), where children were mainly sedentary (71%).
- Access to indoor gyms and integrating opportunities for classroom movement during high-heat days may help preserve children's physical activity opportunities and reduce sedentary time when outdoor recess is restricted.

Physical inactivity is a leading risk factor for developing noncommunicable diseases worldwide.[1] Over 80% of youth globally do not meet the recommended 60 minutes of daily moderate to vigorous physical activity (MVPA),[2] putting them at increased risk for long-term health issues. Physical activity rates in the United States are similar, with 72% to 80% of youth failing to meet physical activity recommendations, resulting in a "D–" grade for the 10th consecutive year on the 2024 US Report Card on Physical Activity for Children and Youth.[3] Increasing physical activity therefore is a public health priority, and focusing on environments that promote movement in places where children spend most of their time, such as schools, is critical. Schools provide both formal and informal opportunities to develop behaviors and knowledge to support lifelong movement. The World Health Organization specifically identifies schools as a target setting for promoting physical activity among children and youth.[4] However, in many regions across the globe, extreme weather events are causing traditional outdoor play opportunities during school to be canceled or modified, impacting children's ability to be physically active.[5]

An established body of literature has demonstrated that children's physical activity levels vary depending on the season, with systematic reviews showing that children and adolescents are less sedentary and more active when the weather is favorable.[6–8] For example, a longitudinal cohort study in Norfolk, United Kingdom, found that increased rainfall decreased physical activity and increased sedentary time among children during the spring and summer months.[9] In Australia, schoolchildren engaged in significantly less MVPA in spring and summer because of higher temperatures.[10] A 5-year longitudinal study in Canada with 1293 adolescents ages 12–13 years showed that physical activity decreased during the winter because of rain and snow by an average of 7%,[11] and a systematic review found that physical activity decreased by 11% for every 10 cm of snowfall.[12] In

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addition to seasonal impacts, extreme weather affects children's physical activity opportunities, as intense events (eg, fires, hurricanes, storm tides, or flooding) and extreme high (ie, heat waves) and low temperatures (ie, cold waves) have a major impact on schools,[13] with 90% of school closures in the United States caused by extreme weather.[14] When school remains in session during extreme weather events, outdoor activity, such as recess, is often canceled or moved indoors, which has implications for children's health.

Outdoor play is important for children's healthy growth and development. Recess is often the only school day opportunity that children can engage in outdoor activities. Traditional outdoor recess is a primary source of children's movement, contributing up to 17% of MVPA[15] and 44%[16] of steps accumulated during the school day. Outdoor play is also one of the few opportunities for children to engage in unstructured play and interact with nature—both of which are crucial for developing social skills, such as cooperation and conflict resolution,[17] as well as cognitive, behavioral, and mental health.[18] When recess is moved indoors, free play is often limited as schools commonly restrict activity to individual classrooms where students have less autonomy to choose what they play and who they play with,[19] likely impacting physical activity as children are more physically active outdoors compared with indoors.[20] An international group of experts released a position statement on the benefits of active outdoor play for children based on the established research showing health benefits of outdoor play.[21] Given the benefits of outdoor play, it is critical to better understand how canceling or moving traditional recess inside impacts children's physical activity.

The decision to move traditional outdoor recess indoors is often made by school administrators who deem outside conditions as unfavorable.[13] However, the lack of sufficient indoor space, resources, and supervision across schools poses challenges to providing equitable opportunities for active play.[19] Although people prefer to stay indoors during extreme weather,[22] US schools often lack available indoor facilities to support movement. While most (75%) US elementary schools in the Northeast and Midwest have gyms, only 60% of schools in the South and 46% of schools in the West have gyms (National Center for Education Statistics—Senior Analyst, email communication, March 4, 2025). Even when indoor facilities (eg, gyms or multipurpose rooms) are available, they are rarely accessible since they are often used for physical education classes and lunch throughout the school day. One study of schools in Maine reported that only two-thirds of schools provided students with an opportunity to be active during indoor recess due to inadequate space.[23] This lack of adequate indoor space is particularly concerning in the southern and western regions of the United States, where 94% of schools start in July or August,[24] the 2 warmest months of the year, with the highest rate of heat-related illness.[25]

The issue of heat is especially pronounced in the Phoenix region of Arizona (AZ), one of the hottest metropolitan areas in the United States, where the typical school year begins in July. During the school months of July to October over the past 5 years in Phoenix, 46 to 92 days (mean = 66 d) exceeded 105 °F (~41 °C).[26] A 2023 study of 61 schools in Phoenix found that up to 7 weeks of recess were disrupted in the first quarter of the school year alone (equivalent to 16% of the academic year).[27] Some schools in the study reported modifications to improve the thermal comfort of students during outdoor recess by adjusting schedules to offer play during cooler morning hours, shortening the amount of time students were outside, or restricting play to shaded areas; however, in most cases (59%), heat caused schools to cancel recess entirely or move it indoors. This decision making is similar to a Canadian qualitative study on the impact of inclement weather on outdoor recess, where teachers and administrators noted adverse weather conditions led to indoor recess over 26 school days, resulting in disruptions to teacher planning time, as well as student social interactions, physical activity, time spent in nature, and increased screen time use.[19]

With extreme weather events increasing in frequency and intensity in many regions of the world,[28] there is a need to understand the impact of disruptions to outdoor play on children's physical activity during school. Further, children's movement during indoor play remains poorly understood, as most research is conducted in outdoor settings. Therefore, this study aimed to assess the amount and intensity of children's physical activity when school recess was indoors (in classrooms or gyms) compared with traditional outdoor recess during periods of high heat. According to the Theory of Expanded, Extended, and Enhanced Opportunities (TEO), physical activity behavior change is positively impacted by 3 types of environmental and contextual modifications: Expansion (eg, offering new opportunities for physical activity), Extension (eg, allotting additional time to physical activity), and Enhancement (eg, increasing physical activity opportunity above routine practice).[29] Based on this theoretical premise, we hypothesized that children's physical activity intensity would be lower during indoor recess time because of unexpanded, or limited, opportunities (decreased space and equipment to support active play).

Method

Research Design

In this cross-sectional observational study, we compared movement intensity levels (MPVA, light physical activity [LPA], and sedentary) among children in fifth grade (N = 251; ages = 10–12 years) during recess at 5 schools in the Phoenix Metropolitan Area (AZ, United States) over 5 weeks between August and September 2024 with typical high heat, considered as temperatures over 90 °F. During the study period, school personnel determined whether to hold recess indoors or outdoors at the start of each day, mostly based on the daily temperature cutoff.

Recruitment and Setting

Schools were randomly selected to participate from a convenience sample of schools in the Phoenix Metropolitan Area that responded to a prior survey (external to this project),[30] indicating that high heat affected outdoor recess. This sample of 57 schools was then stratified by location (either urban or suburban, as approximately 40% of the region's population resides in urban areas),[31] and a random number generator in Excel was used to select a mix of schools from either location group. Principals at selected schools were invited to participate in the study until 5 schools agreed (7 schools declined the invitation initially). Participation included consent for researchers to observe fifth-grade students during recess over 1 week, collect environmental data throughout the schoolyard (eg, air temperature, surface temperature), and survey students about their emotional state following recess. Neither of the latter 2 data sources is reported in this study. All children who participated also received a $5 gift card upon completion of the study. Additionally, gift cards for classroom teachers to administer student surveys ($25) and a $500 equipment bundle for recess per school were offered as incentives.